GPU-accelerated framework for multi-slice strong–strong beam–beam simulation

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ABSTRACT

Strong–strong beam–beam simulation must evolve two bunches self-consistently while controlling field discretization, macroparticle noise, and computational cost. We present Octopus, a Julia/CUDA framework for multi-slice, six-dimensional tracking whose common collision driver supports four transverse field models, from a soft-Gaussian closure to full particle-in-cell (PIC) solvers. Two numerical developments address errors exposed by adaptive PIC: a deposition-consistent Gaussian/residual field split and node-indexed meshes that remove the mesh-mismatch component of slice-boundary discontinuities. An ensemble analysis separates mean-field bias from finite-particle fluctuation and shows that mesh refinement can reduce the former while increasing the latter. A collision-time wavefront schedule runs dependency-free slice pairs concurrently on the GPU while preserving the per-slice collision order exactly. Validation progresses from structural identities and analytic fields to Vlasov theory and the unmodified BeamBeam3D code. At the tested settings, PIC-family and BeamBeam3D round-beam coherent factors lie within 1.42% of the Vlasov reference, and the largest same-setting cross-code difference is 0.31%. A historical 15-slice, 8192-turn EIC application gives comparable emittance evolution under nonidentical inputs. Because complete source and raw-output provenance was not retained, it is qualitative context rather than source-pinned validation. An EIC-scale case with 3.58×10^6 macroparticles advances in about 0.31 s per turn on one NVIDIA RTX 4500 Ada GPU. Together, the error analysis, validation, and timing results support practical strong–strong studies within the tested domain.

1. Introduction

The beam–beam interaction is a fundamental luminosity limit in circular colliders. A weak–strong model freezes one beam as an analytic distribution, whereas a strong–strong model evolves both bunches self-consistently and is therefore required for coherent two-beam modes and mutual emittance evolution. The synchro-beam map of Hirata et al. [1], the strong–strong formulation of Ohmi [2], and the parallel PIC implementation in BeamBeam3D [3, 4] established the numerical basis for multi-slice circular-collider studies [5–8]. Their cost nevertheless remains substantial: every turn couples many slice pairs, each requiring charge deposition, an open-boundary field solve, and two particle gathers.

Octopus is a Julia [9] framework with a CUDA backend [10] for this workload. It combines six-dimensional weak–strong tracking and multi-slice strong–strong collisions in one set of conventions, with four field models behind one collision interface. The production solver uses a fixed number of cells whose physical extent adapts to each interacting pair. Reduced beam–beam meshes [11], Gaussian/residual (δf) decompositions [12, 13], and slice-endpoint potential interpolation [14, 15] are established. The numerical contributions here are narrower: an equal-weight full-f field splitting in which a live Gaussian reference is projected analytically through the charge deposition operator, and the identification of a mesh-consistency failure when adjacent slice pairs construct adaptive transverse grids independently. A node-indexed grid removes this mesh-mismatch term while retaining live-source evolution. On the computational side, a collision-time *wavefront* schedule (Section 6) runs dependency-free slice pairs concurrently on the GPU without changing the per-slice collision order.

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Within the common GPU-oriented driver, the four solvers have distinct production and diagnostic roles. We evaluate them through a connected sequence: field-error decomposition, analytic and BeamBeam3D collective tests, a historical EIC emittance application, and production-scale timing. The source-pinned tests assess accuracy; the EIC application and timing establish the physical and computational scope reached in practice.

Sections 2 and 3 define the physics and algorithms; Sections 4 and 5 analyze error and validate collective observables. Section 6 reports performance, followed by the conclusions.

2. Physics model and conventions

2.1. Coordinates and the coupling constant

Octopus tracks the canonical 6D coordinates $\mathbf{q} = (x, p_x, y, p_y, z, p_z)$, with z the longitudinal position relative to the reference particle (positive toward the bunch head) and $p_z = (P - P_0)/P_0$ the relative momentum deviation. Beams are ultrarelativistic and counter-propagating; each collision is decomposed into thin longitudinal slices, and each slice pair interacts through a two-dimensional transverse potential. All solvers share one coupling constant per kicked beam,

$$k_{\mathrm{bb}} = q_1 q_2 \frac{N r_0}{\gamma}, \tag{1}$$

where $q_{1,2} = \pm 1$ are the charge signs, N the source bunch population, and r_0 and γ the classical radius and Lorentz factor of the *kicked* species. The transverse kick of a Gaussian slice of weight w is $\Delta\mathbf{p}_\perp = k_{\mathrm{bb}}\, w\, \mathbf{K}_{\mathrm{BE}}$, where $\mathbf{K}_{\mathrm{BE}}$ is the Bassetti–Erskine kick function normalized so that $|\mathbf{K}_{\mathrm{BE}}| \to 2/r$ far from a round slice [16]. With this normalization the nominal beam–beam parameter is $\xi_{x,y} = N r_0 \beta^*_{x,y}/[2\pi\gamma\, \sigma_{x,y}(\sigma_x + \sigma_y)]$ with the source beam's sizes, reducing to $\xi = N r_0 \beta^*/(4\pi\gamma\sigma^2)$ for round beams; all ξ values quoted in this paper are nominal (design σ), and the PIC kick is normalized per deposited source macroparticle so that it reduces to $k_{\mathrm{bb}} w\, \mathbf{K}_{\mathrm{BE}}$ for a Gaussian slice.

2.2. The synchro-beam map

A particle at longitudinal position z colliding with a thin source slice at z_* meets it at the collision point $S = (z - z_*)/2$ along the particle's own axis [1]. The collision map is the composition $\mathcal{M} = \mathcal{D}^{-1}\mathcal{B}\,\mathcal{D}$ of a *virtual drift* $\mathcal{D}$ to the collision point, the thin transverse kick $\mathcal{B}$, and the inverse drift, with maps composed *right to left* — the rightmost factor $\mathcal{D}$ acts first. Because the drift length depends on the canonical coordinate z, symplecticity requires a compensating change of p_z — the "slingshot" term. For the default paraxial (Hirata) drift with drift Hamiltonian $H = (p_x^2 + p_y^2)/2$ the energy exchange is

$$\Delta p_z^{\mathrm{sling}} = \frac{(p_x^+)^2 + (p_y^+)^2 - (p_x^-)^2 - (p_y^-)^2}{4}, \tag{2}$$

where superscripts $-$ and $+$ denote values before and after the transverse kick, respectively. Octopus also provides chromatic and exact-Hamiltonian virtual drifts.

For a Gaussian strong source, the analytic longitudinal kick at the collision point generalizes Hirata's formula to a moving source centroid and a fully transported source covariance $A = \left(\begin{smallmatrix} a & b \\ b & d \end{smallmatrix}\right)$, including the transverse coupling $b = \sigma_{xy}$ [17]. Let $u = -S$ denote the collision position measured along the *source* axis, and let $\mathbf{C}(u)$ and $A(u)$ be the source centroid and covariance there. All charge and normalization factors are absorbed into the slice potential U, with transverse kick $\mathbf{F} = -\nabla U$. Since z enters only through S, symplecticity requires $\Delta p_z^{\mathrm{CP}} = -\partial U/\partial z$ evaluated at the collision point, and with $u_z = -1/2$ the Gaussian heat-equation identity $\mathrm{d}U = \frac{1}{2} H_U : \mathrm{d}A$ (with $H_U = \nabla\nabla U$ the transverse Hessian) gives the coordinate-invariant form

$$\Delta p_z^{\mathrm{CP}} = \tfrac{1}{2}\, \mathbf{F}\cdot\mathbf{C}_u + \tfrac{1}{4}\, H_U : A_u, \tag{3}$$

where $\mathbf{C}_u$ and A_u are derivatives with respect to u. The Gaussian heat-equation identity is essential to this analytic form; PIC and spectral routes obtain the corresponding longitudinal kick from numerical potential differences. In principal axes Eq. (3) splits into centroid, principal-size, and *principal-axis rotation* contributions; the rotation term vanishes only for a frozen tilt, so a model that rotates coordinates and forces correctly but holds the tilt fixed omits it. Eight analytic limits and the 6D symplecticity of the composed map are tested in Section 5.

2.3. Source models and longitudinal slicing

The weak–strong source is a full 6D Gaussian specified by its 6×6 covariance. Slices are obtained by *conditional* (not projected) slicing: writing $\mathbf{w} = (x, p_x, y, p_y)$ for the transverse block,

$$\boldsymbol{\mu}_{w|z_i} = \boldsymbol{\mu}_w + \frac{\Sigma_{wz}}{\Sigma_{zz}}(z_i - \mu_z), \qquad \Sigma_{w|z} = \Sigma_{ww} - \frac{\Sigma_{wz}\Sigma_{zw}}{\Sigma_{zz}}, \tag{4}$$

so all delta slices of a joint Gaussian share one transverse covariance and differ only in their 4D centroids. This cleanly separates the two dispersion-like couplings: linear crab dispersion moves slice centroids without changing the conditional slice size, while momentum dispersion widens and couples the within-slice covariance.

In the strong–strong task both beams are live macroparticle ensembles. Each beam is cut into n longitudinal slices (equal-area, normal-quantile, equal-width, or user-specified boundaries); slice pairs interact in collision-time order, and the transverse kick, together with the longitudinal kick obtained from analytic moments or potential differences, is applied to both beams. In the PIC path the field of a source slice is solved at slice-boundary nodes and interpolated linearly (default) or quadratically in z; the Gaussian-subtracted and spectral paths use linear interpolation, whereas soft-Gaussian evaluates each particle directly. In all cases the transverse evaluation point is drift-transported to its collision point. Endpoint interpolation assumes that both sides of a slice boundary represent the nominally shared node with the same discrete transverse operator. Independently adapted per-pair meshes violate that assumption; Section 3.2 describes the node-indexed alternative.

2.4. Machine elements

Between collisions, beams are tracked through linear 6D one-turn maps or element sequences: multi-harmonic thin crab cavities, Hirata's crossing-angle Lorentz boost pair [18], chromaticity kicks, dispersion and coupling maps, and a lumped radiation element applying damping and quantum excitation in decoupled, dispersion-free coordinates. Stochastic excitation uses a counter-based Philox random number generator [19] keyed by $(\mathrm{seed}, \mathrm{turn}, \mathrm{stream}, \mathrm{particle}, \mathrm{component})$, so the excitation is independent of CPU threading and CUDA launch geometry. The crab-crossing collision geometry follows the EIC scheme [20, 21]. Luminosity is accumulated per slice pair as the discrete overlap integral $L_{\mathrm{pair}} \propto (h_x h_y)^{-1} \sum_{ij} q_{ij}^{(1)} q_{ij}^{(2)}$, where $q^{(1,2)}$ are the two slices' charges deposited on a shared bounding grid of spacing (h_x, h_y), normalized by macroparticle weights and scaled by bunch populations and collision frequency — i.e. the standard mesh estimator of $\int \rho_1 \rho_2 \, dx \, dy$ per colliding slice pair. It uses the pre-collision live slice particles transported to the collision midpoint. Soft-Gaussian instead averages one slice's analytic Gaussian density over the other slice's live midpoint sample; it is not a closed-form Gaussian–Gaussian overlap and retains sampling noise and non-Gaussian structure in the sampled slice. No absolute cross-calibration against an external luminosity code is performed; the analytic anchor of Section 5.5 tests a relative geometric factor.

3. Numerical methods

3.1. Four transverse field solvers

The collision driver owns slicing, collision ordering, the longitudinal map, diagnostics, and backend dispatch. A solver supplies the transverse field and potential through a common interface, so the same physical case can be run with four numerical models:

- **PIC** is the production model. Cloud-in-cell (CIC) or triangular-shaped cloud (TSC) charge deposition is followed by a zero-padded Hockney convolution for the open-boundary two-dimensional Poisson problem [22]. The logarithmic Green function is integrated over each source cell [4, 23]; transverse fields are obtained with second- or fourth-order differences and gathered with the deposition stencil. Each slice pair uses a fixed $n_x \times n_y$ cell count over the particle bounding box plus 1.5 cells on each side (three cells total per axis). By default, the first grid for each directed slice pair is enlarged by 25% and cached; it is rebuilt when the live domain no longer fits or becomes less than half the cached width or height.

- **Soft-Gaussian** replaces each live slice by the Gaussian having its measured centroid and marginal rms sizes, then applies the Bassetti–Erskine field [16]. This uncoupled hot path is the default used here. The optional `include_sigma_xy=true` route measures the full transverse covariance, evaluates the field in its instantaneous principal axes, and includes the rotation derivative in the longitudinal kick. The solver is grid-free and fast, but it is a moment closure rather than a representation of distribution-shape evolution.

- **Gaussian-subtracted PIC** is an experimental research path that uses the same full-f, equal-weight particles as PIC. For each directed slice interaction, it recomputes a Gaussian reference $G_{\hat{\theta}}$, subtracts its analytically expected nodal deposit from the particle deposit, solves the residual, and adds the analytic Gaussian field:

$$\mathbf{K}_h^{\mathrm{GS}}(\mathbf{x}) = \mathcal{I}_h \mathcal{P}_h \left[\mathcal{D}_h f_{N_p} - \alpha_h \mathcal{D}_h G_{\hat{\theta}}\right](\mathbf{x}) + \mathbf{K}_{\mathrm{BE}}(\mathbf{x}; G_{\hat{\theta}}), \qquad \alpha_h = \frac{\sum_{ij} (\mathcal{D}_h f_{N_p})_{ij}}{\sum_{ij} (\mathcal{D}_h G_{\hat{\theta}})_{ij}}. \tag{5}$$

  Here $\mathcal{D}_h$, $\mathcal{P}_h$, and $\mathcal{I}_h$ denote unit-charge deposition, the mesh Poisson solve, and field gathering, and f_{N_p} is the empirical density of the N_p deposited source macroparticles; the implementation stores particle counts and applies the equivalent common normalization in the kick scale. The analytic add-back has unit charge and is not multiplied by α_h. Thus, even for an exact matching discrete operator, neutralization changes the nominal full-f split by $(1 - \alpha_h)\mathbf{K}_{\mathrm{BE}}$. For six representative sources, the default five-sigma margin gives $|1 - \alpha_h| \leq 2.09 \times 10^{-7}$; with no enforced margin the largest value is 6.80×10^{-3}. The neutralization is therefore a controlled finite-box approximation, not an exact algebraic identity. The one-dimensional expectations of the CIC or TSC weights are evaluated analytically. The default $G_{\hat{\theta}}$ is separable in the laboratory grid axes and matches the centroid and marginal rms sizes; any transverse correlation remains in the PIC residual. The factor α_h makes the finite-box residual charge-neutral before the FFT. A second-order conditional expansion can represent a correlated reference, but the present accuracy study is restricted to the uncoupled case. The method is related to nonlinear δf beam–beam simulation [12, 13], but it neither evolves residual particle weights nor replaces the full-f sample. Its purpose here is to correct the mesh treatment of a smooth near-Gaussian component, not to assert general sampling-variance reduction. The comparisons below use the uncoupled CIC, linear-interpolation, slice-pair configuration listed in Table 1.

- **Spectral** expands the charge and potential in a double sine series on a Dirichlet box and differentiates the potential spectrally. Its boundary condition and discretization are independent of the free-space PIC path, making it an independent field-solver cross-check rather than a production solver.

The four models consequently have different roles. PIC retains arbitrary distribution structure, soft-Gaussian supplies a rapid moment-level diagnostic, Gaussian subtraction lowers the near-Gaussian discretization floor, and the spectral solve exposes errors shared by neither FFT Green functions nor finite-difference fields. Their quantitative comparison is given in Section 4.

3.2. Node-indexed interaction meshes

Reduced or adaptive-extent meshes concentrate resolution on a beam core [11], while endpoint-potential interpolation is an established way to avoid longitudinal slice-boundary jumps [14, 15]. The continuity argument assumes that the right endpoint of slice s and the left endpoint of slice $s+1$ are represented by the same discrete transverse operator. If the two slice pairs choose their particle-bounding meshes independently, the nominally shared physical field is discretized on different domains. Its truncation error then becomes a step in z, giving a relative transverse-kick jump of order 10^{-3} in the test of Section 4.3.

Octopus can instead key the mesh by the longitudinal interaction node. Both sides of a boundary then use the same node-indexed mesh. The live field itself is recomputed at each scheduled encounter because the source may have been kicked. The longitudinal kick is a difference of nearby potentials and requires

Table 1
Solver roles and settings used in the single-kick field comparisons of Section 4. "Uncoupled" means that σ_{xy} is not included in the analytic reference; correlated structure remains in the particle distribution.

solver	source representation	role in this work	evaluated setting
PIC	arbitrary full-f particles	production and mesh/interpolation studies	CIC; cell-integrated Green function; second-order field difference; linear z; pair mesh
soft-Gaussian	live moments	fast moment-closure diagnostic	uncoupled default
Gaussian-subtracted PIC	full-f particles plus Gaussian reference	experimental near-Gaussian field study	CIC, uncoupled, linear z, pair mesh, 64^2
spectral	arbitrary full-f particles	independent Dirichlet field check	linear, independent field check

correlated discretization error; each directed interaction therefore solves the left and right endpoints plus the right endpoint on the left-node mesh. With the longitudinal kick enabled, the current implementation evaluates $3N_{\mathrm{int}}$ planes for N_{int} directed interactions, versus $2N_{\mathrm{int}}$ for pair-indexed grids. A frozen-source construction has only $2N_{\mathrm{int}} + 1$ algebraically distinct node planes, but a live source is kicked between uses, so reusing a plane would make the later field stale. Thus $3N_{\mathrm{int}}$ is required by the present self-consistent update order rather than merely being an unimplemented cache optimization. Absent a demonstrated multi-turn benefit, the faster slice-pair grid remains the production default. An experimental source-slice-shared grid also removes the frozen-source mesh-mismatch term, but can coarsen cells when the interaction windows have unequal extents and is not treated as a universal remedy.

3.3. Longitudinal interpolation order

A second discontinuity arises because two-node interpolation assigns a constant potential slope within each slice. A three-node option adds the slice midpoint and uses a quadratic interpolant. On a common transverse grid, neighboring interpolants share the boundary value; for equal slice widths their mismatch decreases from first to third order in the width. TSC also makes charge deposition smoother under source drift than CIC. In a frozen scan of an EIC-like flat beam (seven quantile slices, 64^2 mesh, 2×10^5 macroparticles), quadratic interpolation plus TSC reduced the largest normalized longitudinal-kick jump from 55% to 9.5×10^{-4}. It requires a third plane per interaction. This is a field-continuity result; no claim is made here that it changes a long-term observable.

4. Error budget

Three effects must be separated to interpret a strong–strong PIC result: deterministic field error, finite-particle fluctuation, and discontinuities introduced by changing interaction grids. Treating them separately connects solver design to the collective validation in Section 5.

4.1. Total single-kick error

For a known Gaussian source we evaluate the numerical kick on an 81×81 grid spanning $\pm 4\sigma$ in both planes and normalize errors by the maximum exact Bassetti–Erskine kick on that grid. Figure 1 shows the median pointwise error versus macroparticles per slice. At 10^4 particles, all three mesh solvers are dominated by source sampling: changing the field model matters less than adding particles. At 10^6, the deterministic floor becomes visible. On the paired 400^2 normal-quantile source, the evaluated configurations give median errors of 1.57×10^{-4} (round) and 2.38×10^{-4} (flat) for 64^2 Gaussian-subtracted PIC, versus 4.07×10^{-4} and 9.66×10^{-4} for 128^2 PIC. Perturbation tests limit the scope of this ordering: for a 30% narrow component the hybrid 95th-percentile error exceeds that of 128^2 PIC, and for a coupled offset mixture its median and rms errors exceed those of 64^2 PIC. The demonstrated use case is therefore axis-aligned and near Gaussian, not general superiority.

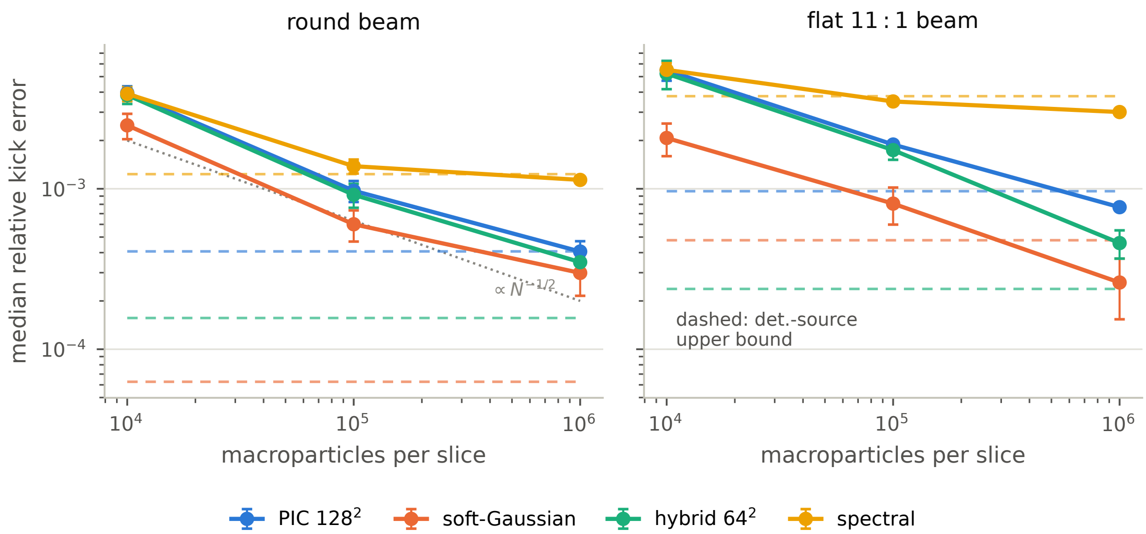

Figure 1: Median transverse-kick error relative to the peak exact kick for a round source and an 11:1 flat source, evaluated at 81^2 probes over $\pm 4\sigma$. Symbols and bars are means and sample standard deviations over six independent batch medians; each batch contains five paired source draws at 10^4 and 10^5 macroparticles and three at 10^6. Dashed lines are deterministic-source estimates from a 400^2 Gaussian quantile lattice and contain source-quadrature as well as solver error; they are not bounds. PIC uses 128^2 cells, Gaussian-subtracted PIC 64^2, and the independent spectral grids are 127^2 (round) and 127×383 (flat), with domain half-width factors 16 and 8 relative to the larger measured source rms.

4.2. Separating systematic and fluctuating components

A total-error statistic does not distinguish mesh bias from the sampling noise transmitted by the solver. Let $\mathbf{e}_i^{(r)} = \mathbf{K}_{h,i}^{(r)} - \mathbf{K}_i^{\text{exact}}$ be the kick error at field point i for independent source realization r. For R realizations we define

$$B_h = \frac{\text{median}_i \, \|\bar{\mathbf{e}}_i\|}{K_{\max}}, \qquad S_h = \frac{\text{median}_i \sqrt{s^2(e_{x,i}) + s^2(e_{y,i})}}{K_{\max}}, \tag{6}$$

where the overbar and s are the sample mean and standard deviation over r, respectively. B_h estimates the systematic field error and S_h the realization-to-realization fluctuation. A centered Rademacher sign-flip bootstrap estimates the finite-R sampling floor of B_h without imposing a Gaussian or isotropic model; its null assumes sign symmetry and exchangeability of the centered realization errors.

Table 2 first summarizes 19 independent blocks, each containing $R = 100$ sources with 10^5 macroparticles. Changing CIC from 64^2 to 256^2 gives descriptive ratios of block-mean B_h values of 4.02 (round) and 6.70 (flat), while S_h increases by 9.51% and 18.47%. Per-block ratios of fine-grid B_h to the median of its own centered sign-flip null distribution span 0.71–1.45 (round) and 0.73–1.34 (flat): the 256^2 bias sits at the sampling floor of its own estimator, so the finite-R floor inflates the measured B_{256} and the descriptive ratios 4.02 and 6.70 are lower bounds on the true $64^2 \to 256^2$ bias reduction. Because the norm and spatial median are applied after each 100-source average, these block statistics contain a positive finite-R magnitude floor.

The second panel therefore pools all 1900 source fields before evaluating Eq. (6). A delete-one-block normal interval summarizes sensitivity to the 100-source blocks; because B_h is nonnegative and has a finite-R norm floor, exclusion of zero by this interval is not a zero-bias test. The null comparison instead applies one Rademacher sign to each full realization vector, preserving dependence across field points. For the round source, $B_{256} = 3.507 \times 10^{-5}$ lies below the sign-flip null 95th percentile 3.803×10^{-5}. For the flat source, $B_{256} = 7.138 \times 10^{-5}$ exceeds the corresponding 95th percentile 6.589×10^{-5}. Thus the pooled fine-grid bias is unresolved for the round source but resolved for the flat source. Two grid sizes still establish neither a directional bound nor a convergence rate. Resolution, deposition order, and particle count control different parts of the error budget.

Table 2
PIC error decomposition for 10^5 macroparticles and CIC deposition. Panel A gives means $\pm$ sample SD over 19 independent $R = 100$ blocks; bootstrap 95% intervals for the displayed ratios are [3.73, 4.33] and [6.19, 7.26], quantifying block-sampling variability of these floor-limited descriptive ratios, not the true bias reduction. Panel B pools all $R = 1900$ source fields; entries are in units of 10^{-4}. Delete-one-block 95% intervals for round/flat B_{256} are [0.263, 0.438]/[0.462, 0.965], and those for S_{256} are [14.641, 14.969]/[29.844, 30.536], in the same units.

A: separate $R = 100$ block statistics				
source	B_{64}	B_{256}	B_{64}/B_{256}	$(S_{256}/S_{64} - 1)$
round	$(4.694 \pm 0.136) \times 10^{-4}$	$(1.167 \pm 0.211) \times 10^{-4}$	4.022	$(9.513 \pm 0.372)\%$
flat 11:1	$(1.398 \pm 0.096) \times 10^{-3}$	$(2.087 \pm 0.381) \times 10^{-4}$	6.696	$(18.468 \pm 1.345)\%$
B: pooled $R = 1900$ statistics ($\times 10^{-4}$)				
source	B_{64}	B_{256}	S_{64}	S_{256}
round	4.494	0.351	13.513	14.805
flat 11:1	14.070	0.714	25.430	30.190

4.3. Measured slice-boundary discontinuity

In the frozen-field test of Section 3.3, changing from slice-pair to node-indexed grids reduces the relative transverse-kick jump from order 10^{-3} to 1.1–5.0×10^{-9} horizontally and 2.2–3.1×10^{-8} vertically (Fig. 2a). Common and source-slice-shared grids reach the same level, confirming that mesh identity across the boundary, rather than node indexing by itself, produces the cancellation. When live-source evolution is restored, the node-indexed mesh still removes the mesh-switch term but recomputation after the source kick leaves jumps of 2.2–3.6×10^{-5} horizontally and 5.6–8.1×10^{-5} vertically. Panel (b) separately shows the longitudinal interpolation test discussed above. These measurements isolate numerical mechanisms; they do not establish the size of any multi-turn bias.

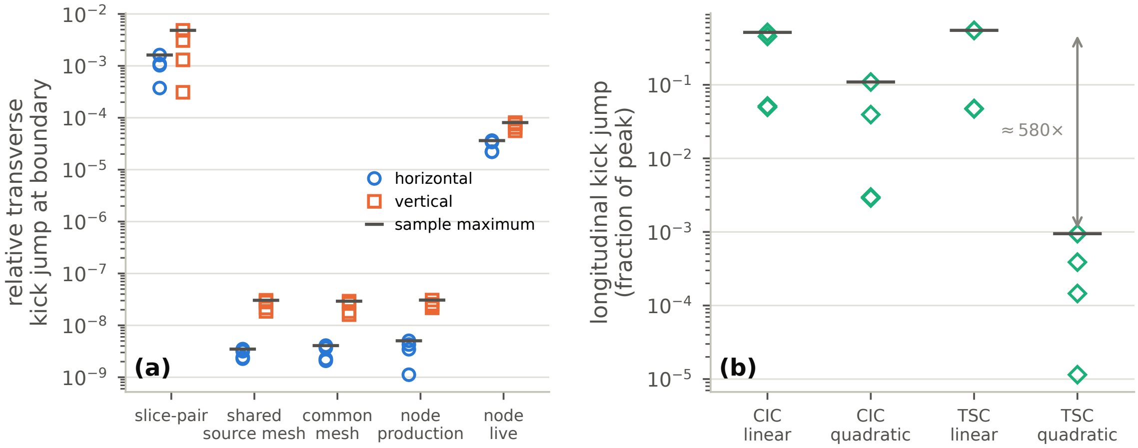

Figure 2: Slice-boundary tests for a flat EIC-like source, seven quantile slices, 64^2 cells, and 2×10^5 macroparticles. Each jump is divided by the peak magnitude of the corresponding exact kick over the scan. (a) Four frozen-source grid constructions and a node-indexed live-source control; each marker is one of two source slices at one of two central boundaries. The "node production" values use the full scheduled node calculation. (b) Frozen-source longitudinal jumps for four deposition/interpolation combinations. Diamonds are the same four samples and horizontal bars mark their maxima.

4.4. Interpretation

The decomposition explains why a finer mesh need not produce a quieter multi-turn calculation: it resolves a more accurate mean field and also more of the finite-particle structure. It does *not* by itself predict emittance growth, diffusion, or a required particle count, which depend on temporal correlations, resampling, damping, and lattice dynamics. We therefore use Eq. (6) as a solver diagnostic and validate multi-turn observables independently in Section 5.

5. Validation of the collective physics

No single comparison can establish the correctness of a strong–strong program, so the validation is deliberately layered. Structural identities test map conventions; analytic Gaussian fields and the independent spectral solver test transverse kicks; ensemble and mesh studies test numerical error; and CPU/CUDA comparisons test backend consistency. Vlasov theory and the unmodified BeamBeam3D code then test collective dynamics. A historical EIC case illustrates the complete application chain but is not counted as source-pinned validation because its full raw provenance was not retained. AI tools assisted implementation and analysis; their output is not treated as evidence. Each scientific claim is instead tied to an analytic limit, an independent implementation, a convergence or ensemble test, or recomputation from archived numerical records.

Unless stated otherwise, the Octopus PIC comparisons in this section use Float64, CIC deposition, the integrated Green function, second-order field differences, linear longitudinal interpolation, pair-indexed extrema grids, and centroid-centered normal-quantile slicing. The luminosity anchor instead sets $k_{\mathrm{bb}} = 0$ and specifies its CIC luminosity grid independently.

Short CPU/CUDA regression tests provide a bounded backend check. Two-turn soft-Gaussian and standard-PIC cases with 1024 macroparticles per beam, three slices, and (for PIC) a 32^2 mesh passed coordinate and luminosity relative tolerances of 10^{-10}. Eight one-collision Gaussian-subtracted cases with 6000 macroparticles per beam, five slices, and a 64^2 mesh passed the coordinate vector-norm and luminosity criteria at relative tolerance 10^{-9} and coordinate absolute tolerance 10^{-18}; six also passed the stricter elementwise coordinate bound. These are short regression tests, not a claim of bitwise or long-term backend equivalence.

5.1. Structural checks

Structural checks establish the map conventions before collective observables are compared. Decreasing the centered-difference step from 3×10^{-7} to 3×10^{-8} lowers the symplectic defects of the thin synchro-beam and analytic Gaussian maps from $7.91\times10^{-8}/4.24\times10^{-8}$ to $7.90\times10^{-10}/4.24\times10^{-10}$, consistent with second-order finite-difference truncation. The crossing boost and inverse compose to identity to 8.3×10^{-19}, and the strong–strong models approach their frozen-source limits as source energy increases.

5.2. Coherent modes and the Yokoya factor

The coherent σ/π dipole spectrum is a direct test of self-consistent distribution dynamics [24–26]. Two identical round beams collide at one IP with $\xi = 0.005$, tunes $(0.31, 0.32)$, one longitudinal slice, and no crossing angle. One beam starts at 0.1σ offset. We extract the sum and difference mode tunes from 8192-turn Hann-windowed centroid spectra using parabolic peak interpolation and form

$$\Lambda = \frac{Q_\pi - Q_\sigma}{\xi}. \tag{7}$$

A rigid-bunch model gives $\Lambda = 1$; the linearized Vlasov result is about 1.2144 for round Gaussian beams [7, 25].

At the stated 128^2, 0.1σ, 8192-turn setting, three PIC runs give $\Lambda = (1.1973\pm0.0035, 1.2061\pm0.0004)$, where the uncertainties are seed scatter only. Figure 3 shows the centroid spectra and the fitted Q_σ and Q_π for one representative seed. One-at-a-time changes, averaged over the same three seeds, quantify the larger numerical envelope: $128^2 \to 256^2$ shifts (Λ_x, Λ_y) by $(+0.0045, +0.0030)$, $0.1\sigma \to 0.025\sigma$ by $(+0.0070, +0.0060)$, and 8192 to 16384 turns by $(-0.0002, -0.0031)$. A zero-padded diagnostic attributes much of the last vertical shift to three-bin interpolation and FFT-bin phase. Thus the robust claim is a round-beam factor near 1.20–1.21, not a three-decimal continuum limit. Gaussian-subtracted PIC gives the value listed in Table 3, while soft-Gaussian gives $\Lambda = (1.096, 1.100)$: evolving moments recover part of the shift above the rigid value but omit distribution-shape feedback [27].

An exploratory flat-beam calculation has fixed $\xi_x = 0.005$ and therefore $\xi_y = 0.0556$; it is not used as validation of the linear flat-beam limit.

5.3. Cross-code comparison with BeamBeam3D

The same physical configuration — round beams, one IP, single slice, no crossing angle, 10^5 particles, 128×128 transverse grid, 8192 turns — was run without source modification in BeamBeam3D [3] (V1.0, commit `50d01d8`), compiled with GNU Fortran 11.5.0 and OpenMPI 4.1.1. Independent builds and full reruns reproduced the reported centroid outputs. The executable logs a realized single-slice strength of 0.0049932095. Table 3 uses the predeclared nominal denominator $\xi = 0.005$; using the logged value would give $(\Lambda_x, \Lambda_y) = (1.1991, 1.2115)$ instead of $(1.1974, 1.2098)$, a 0.136% normalization change. With the nominal convention, the differences from the three-seed Octopus mean are 0.00016 in x and 0.00371 in y, and less than 4×10^{-6} in Q_σ. The codes share no source but implement the same CIC/integrated-Green method.

Table 3
Fixed-setting round-beam Yokoya factors $\Lambda = (Q_\pi - Q_\sigma)/\xi$ (8192 turns, 10^5 macroparticles/beam, $\xi = 0.005$, one slice). PIC reports mean ± seed SD for three runs, not total numerical uncertainty; the soft-Gaussian and Gaussian-subtracted rows are three-seed means and BeamBeam3D is a single run. BeamBeam3D uses the nominal denominator. The Vlasov reference is 1.2144.

solver	Λ_x	Λ_y	assessment
soft-Gaussian	1.096	1.100	below Vlasov value (closure)
PIC 128^2	1.1973 ± 0.0035	1.2061 ± 0.0004	fixed setting
Gaussian-subtracted PIC 64^2	1.198	1.206	max. distance 1.34%
BeamBeam3D	1.1974	1.2098	independent code

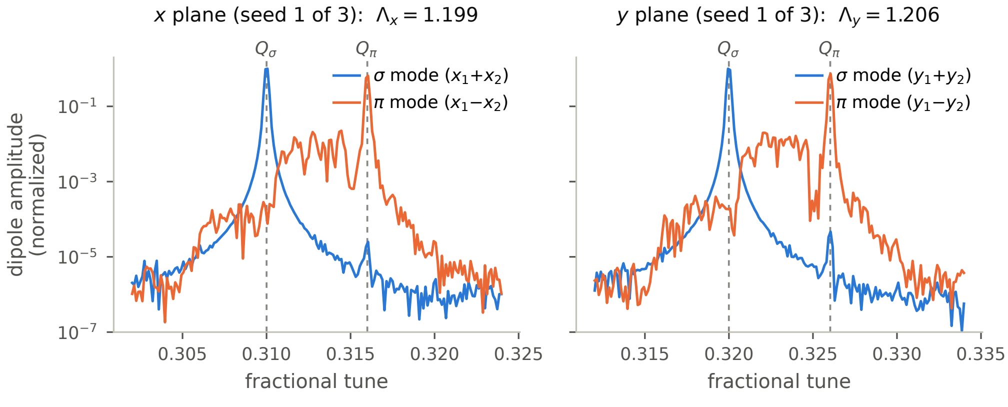

Figure 3: Round-beam PIC centroid spectra for one of three seeds (8192 turns, 10^5 macroparticles/beam, $\xi = 0.005$, one slice). Each plane is normalized to its largest sum- or difference-mode amplitude. Dashed lines mark the fitted Q_σ and Q_π; the other seeds give the scatter in Table 3.

5.4. Multi-slice cross-code benchmark

A five-slice benchmark exercises the synchro-beam map and longitudinal kick. Both codes use $\sigma_z/\beta^* = 0.5$, $|Q_s| = 0.01$, $\xi = 0.005$, 5×10^4 macroparticles per beam, and 4096 turns. Synchrotron modulation produces the line structure shown in Fig. 4. With the displayed symmetric Hann window, the estimator searches $[Q_0 - 0.004, Q_0 + 0.010]$ and refines local maxima above 1% of the in-band maximum by a three-bin parabola. Under that prescription, the noise-excited Octopus and coherently kicked BeamBeam3D horizontal traces each contain six peaks. Frequency-rank pairing gives median/maximum $|\Delta Q| = (1.20, 3.84) \times 10^{-4}$, with the maximum equal to 1.6 raw FFT bins. Varying the peak threshold or window changes the accepted peak count, and the two traces use different excitation protocols. Figure 4 therefore establishes similar horizontal spectral structure, not one-to-one mode identity.

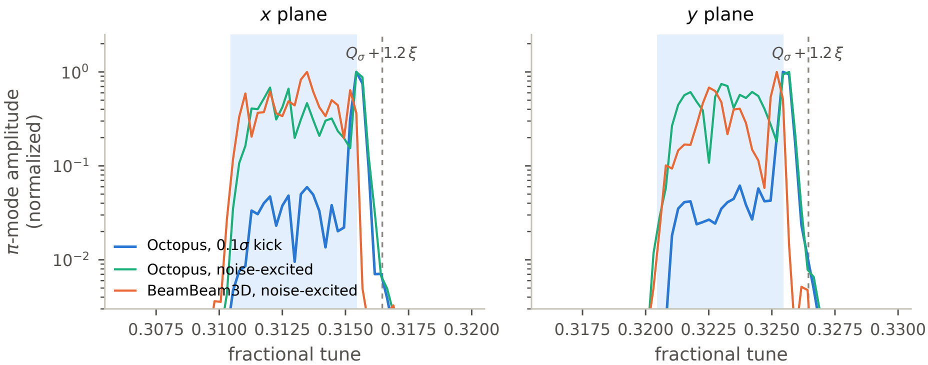

Figure 4: Five-slice normalized difference-mode spectra (5×10^4 macroparticles/beam, $\sigma_z/\beta^* = 0.5$, $|Q_s| = 0.01$, $\xi = 0.005$, 4096 turns). Shading spans $[Q_\sigma, Q_\sigma + \xi]$ and the dashed line marks $Q_\sigma + 1.2\xi$. The noise-excited Octopus and coherently kicked BeamBeam3D horizontal traces each contain six peaks under the stated estimator; the comparison supports similar structure, not common mode identities.

5.5. Crossing-angle and crab-cavity luminosity anchor

For equal frozen round Gaussians ($\sigma_x = \sigma_y = 100\,\mu$m, $\sigma_z = 2$ cm, $\beta^* = 0.5$ m), 15 normal-quantile slices, and 2×10^5 macroparticles per beam, a half crossing angle of $\theta_c/2 = 12.5$ mrad gives Piwinski parameter $\phi = (\sigma_z/\sigma_x)\tan(\theta_c/2) = 2.5$. The analytic luminosity factor is $R_\phi = (1 + \phi^2)^{-1/2} = 0.371391$ [28]. Five paired particle seeds give $L/L_0 = 0.371950 \pm 0.000272$ (sample standard deviation), while ideal crab cavities restore L/L_0 to 1.000000 ± 0.000010. A one-seed scan over 1, 7, 15, and 31 slices and $128^2/256^2$ grids spans 0.37147–0.37195, with no monotonic slice trend. The production estimator retains particle midpoint coordinates; a centroid-only 15-slice quadrature gives 0.369274 and is not its finite-slice target. This checks relative boost/crab/luminosity geometry, not absolute luminosity.

5.6. Historical EIC emittance application

The preceding tests isolate field and collective-mode behavior. To exercise the complete multi-turn chain, we retain one historical Octopus and one historical BeamBeam3D run for a head-on, 15-slice collision at parameters adapted from the EIC design [29]: 10 GeV electrons ($\xi_e = (0.088, 0.100)$, $\beta^* = (0.55, 0.056)$ m, $\sigma^* = (106, 9.5)\,\mu$m) against 275 GeV protons ($\xi_p = (0.0094, 0.0094)$, $\beta^* = (0.80, 0.072)$ m, $\sigma^* = (95, 8.5)\,\mu$m), a 128^2 mesh, 10^5 macroparticles per beam, and 8192 turns. The proton side uses the design optics; the electron side departs from it ($\beta_x^* = 0.55$ versus 0.45 m, $\epsilon_y = 1.61$ versus 1.3 nm), so the spot sizes are unequal and $\xi_{e,x}$ and ξ_p differ from the design values $\xi_e = (0.072, 0.10)$, $\xi_p = (0.012, 0.012)$. The electron damping times are shortened from $(4000, 4000, 2000)$ to $(400, 400, 200)$ turns in (x, y, z) so that the run approaches an equilibrium; the proton remains undamped. Octopus uses electron/proton populations $1.7203 \times 10^{11}/0.6881 \times 10^{11}$, whereas the archived BeamBeam3D decks use $1.72 \times 10^{11}/0.69 \times 10^{11}$. Octopus also uses Gaussian Philox excitation while BeamBeam3D uses variance-matched uniform excitation. The transverse tunes are $(0.08, 0.14)$ for electrons and $(0.228, 0.210)$ for protons in both codes, but Octopus uses synchrotron tunes $(-0.069, -0.01)$ for the electron/proton beams while BeamBeam3D uses $(+0.069, +0.01)$ under the same linear-map sign convention. Initial macroparticles are generated independently, and each code uses its native longitudinal slicing and adaptive field domains. The case therefore tests how the two program chains behave under related, but not matched, inputs.

Table 4
Historical EIC emittance application (15 slices, 8192 turns, 10^5 macroparticles/beam; one run per code under nonidentical inputs). Values are late-time means over common sampled turns 6144–8176 (128 records). Electron values are relative to the design emittance; undamped proton values are relative to each run's first record.

beam	plane	Octopus	BeamBeam3D	difference
electron: equilibrium relative to design emittance				
electron	x	$+11.03\%$	$+10.76\%$	$+0.27$ pp
electron	y	$+4.66\%$	$+3.56\%$	$+1.10$ pp
proton: growth relative to each code's initial value				
proton	x	$+0.576\%$	$+0.643\%$	-0.067 pp
proton	y	$+1.092\%$	$+1.182\%$	-0.090 pp

These trajectories predate the frozen source-state record used elsewhere in this article. The BeamBeam3D decks, log, and raw outputs and the reduced Octopus emittance series survive, but the original executable-to-trajectory identity is incomplete; the native Octopus HDF5 file, converter, command, log, environment, and source identity were not retained. Table 4 and Fig. 5 are therefore a transparently historical, qualitative application comparison, not source-pinned validation or an exactly replayable result.

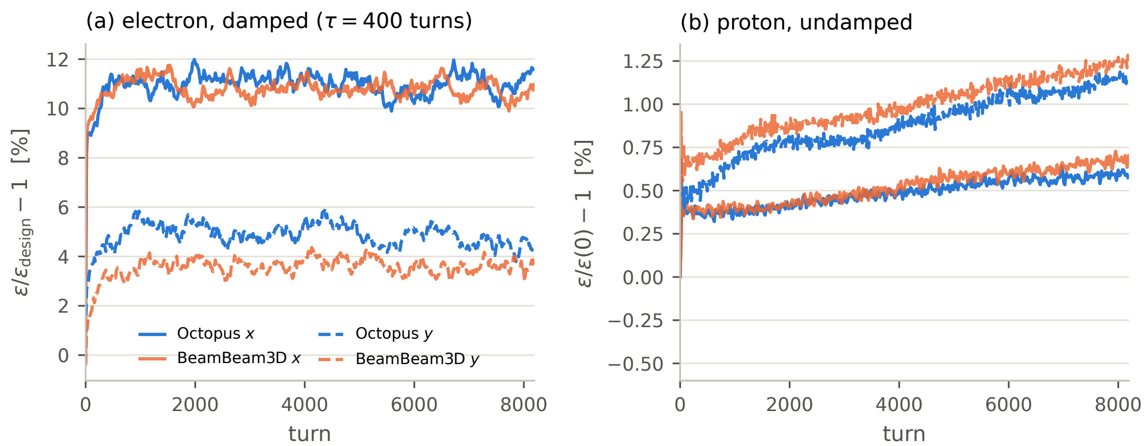

Figure 5: Emittance evolution in the historical EIC application (10^5 macroparticles/beam, 15 slices, 128^2 mesh). Electron curves are relative to design emittance and include shortened radiation damping and excitation; undamped proton curves are relative to each code's initial sample. Blue and orange denote Octopus and BeamBeam3D; solid and dashed lines denote the x and y planes.

The late-time electron means differ by 0.27 percentage points (pp) in x and 1.10 pp in y; proton growth differs by 0.067 and 0.090 pp. These retained curves exhibit the same qualitative evolution and comparable late-time changes in this demanding case. Because there is one realization per code, the stated input differences, and incomplete run provenance, the comparison is physical application context, not validation of identical dynamics or a measure of statistical cross-code uncertainty; the 1.10 pp electron-y difference is about 31% of BeamBeam3D's induced equilibrium excess.

Taken together, the source-pinned tests span map structure, field accuracy, coherent modes, multi-slice spectra, and crossing geometry; the historical EIC application adds long-term emittance evolution without extending that validation claim. Matched ensemble-level code comparison, the linear wide-plane flat-beam limit, and the crab-compensated coherent-mode family of Ohmi et al. [30] remain outside the present study.

6. Performance

6.1. CUDA execution strategy

The multi-slice collision schedule is sequential per slice: each encounter modifies both participating slices, so a slice's later encounters depend on its earlier ones, and one slice pair at a time exposes too little parallelism for the device: a single 128^2 deposit/solve/gather leaves most of it idle. Octopus extracts the concurrency that the physics does permit. Each pair (i, j) is assigned the collision time implied by its slice centers, $t_{ij} \propto -(z_i^{(1)} + z_j^{(2)})$ with both coordinates positive toward each bunch's head, so head–head pairs collide first, and all n^2 pairs are sorted by t_{ij}. A pair is *ready* once every earlier pair sharing either of its slices has completed; the scheduler repeatedly gathers the ready pairs that share no slice into one *wavefront*. Pairs within a wavefront touch disjoint particle sets, so executing the whole batch concurrently reproduces every slice's encounter sequence — and therefore the sequential collision physics — exactly; batching changes only the floating-point reduction order. For n equal-count slices per beam the schedule is the $2n-1$ anti-diagonals of the $n \times n$ interaction matrix: the production 15-slice case runs its 225 pairs in 29 wavefronts of up to 15 concurrent pairs. In the parallel PIC implementation of Qiang et al. [3], parallelism comes from distributing particles and field domains across processes while slice pairs advance in collision-time order; the wavefront schedule is complementary, exposing the concurrency between dependency-free pairs within the schedule itself.

The wavefront then drives the whole CUDA pipeline. Kernels access particles through slice index lists, without temporary gather/scatter copies; the four source-boundary charge and force planes of every pair in a b-pair wavefront are solved in one batched FFT stack of $4b$ planes; persistent workspaces avoid per-pair allocation and kernel construction. Other high-volume operations follow the wavefront: fused two-level reductions set source and target bounds, both longitudinal planes are deposited together, and normalization, spectral multiplication, finite differences, and gathering remain on the device. Scheduling and data movement change, but both backends implement the same ordered collision algorithm. Floating-point reduction order is backend-dependent, so agreement is tested with stated tolerances and is not claimed to be bitwise identity.

The adaptive interaction domains make the Green function itself a recurring cost. The open-boundary solve uses the shifted integrated Green function [4, 23] on a per-pair domain with a fixed cell count. Collider parameters approach the hourglass regime (β^* comparable to σ_z), where head and tail slices are transversely wider than the waist; a single uniform grid sized for the widest pair would concentrate the core slices in a small fraction of the mesh, and the cell size sets the interpolation error exactly where most of the charge sits. Per-pair adaptive domains keep the cells matched to each interaction, but the Green-function FFT then depends on the domain and would otherwise be rebuilt for every directed pair on every turn. Octopus therefore caches two Green FFTs per slice pair, one per kick direction, and reuses them while the pair's live domains still fit. Two solver parameters control the cache: a minimum-ratio threshold declares a cached grid too large once the requested width or height falls below that fraction of the cached one, and a growth margin sets the fractional enlargement applied when an entry is (re)built. The two parameters trade reuse against cell tightness: a small ratio and a generous margin keep entries alive across turns at the price of transiently wider cells, while values near the exact fit track the live domain and rebuild whenever it leaves the tolerance band.

6.2. Benchmark and scaling

The benchmark uses 2.56 million electron and 1.024 million proton macroparticles, 15 quantile slices per beam (225 slice pairs and 450 directed field evaluations), a 128^2 CIC mesh, integrated Green function, linear longitudinal interpolation, pair-indexed adaptive grids, and the EIC crab-crossing lattice [29]. The timed turn includes luminosity output every turn and two HDF5 moment observers. FP64 measurements use the source commit identified below on an NVIDIA RTX 4500 Ada (24 GB, driver 580.119.02), CUDA.jl 6.2.1, Julia 1.12.4, and a single host Julia thread on an Intel Xeon Gold 6430.

At the nominal point, five process means give 0.3129 ± 0.0071 s/turn; the mean of the corresponding process medians is 0.3067 ± 0.0070 s/turn. Multiplying the process-mean 0.3129 s/turn by 10^5 gives an arithmetic extrapolation of 8.69 GPU-hours. The robust process-median scaling values are 0.2048 and 0.4786 s/turn when both particle populations are halved or doubled, and 0.2612 and 0.5106 s/turn for nominal 64^2 and 256^2 meshes (Table 5). A separate idle Julia CUDA context occupied about 3.2 GiB during the measurements but showed zero GPU utilization. The results are elapsed times on one shared-host device, not a multi-architecture performance model. They also carry over to data-center hardware: the weak–strong

Table 5
Measured GPU wall time per turn, including the observers described in the text. Entries are means $\pm$ sample SD across independent process medians after warm-up (five processes for the nominal case and three for the other cases). Scaling windows include one HDF5 observer flush.

configuration	total particles	mesh	median s/turn
half population	1.79×10^6	128^2	0.2048 ± 0.0009
nominal, coarse mesh	3.58×10^6	64^2	0.2612 ± 0.0007
nominal	3.58×10^6	128^2	0.3067 ± 0.0070
nominal, fine mesh	3.58×10^6	256^2	0.5106 ± 0.0019
double population	7.17×10^6	128^2	0.4786 ± 0.0093

companion case at 1.02×10^6 macroparticles, with the same every-turn luminosity and moment output, measures 38 ms/turn on this device (median of five 100-turn windows) and 3.9 ms/turn on an NVIDIA A100 of the NERSC Perlmutter system, a gap that largely reflects the A100's FP64 pipeline running at 1:2 of its FP32 rate, versus 1:64 on this workstation-class GPU, with the A100's higher memory bandwidth also contributing. The nominal strong–strong case runs below 0.2 s/turn on Perlmutter's A100 nodes, consistent with the same FP64-throughput advantage.

7. Summary

Octopus combines four complementary transverse field models, six-dimensional multi-slice collision physics, and wavefront-scheduled GPU execution in one strong–strong framework. The error analysis shows why accuracy cannot be described by mesh size alone: refinement reduces mean-field bias but can expose more finite-particle fluctuation. Assignment-consistent Gaussian subtraction lowers the near-Gaussian discretization error, while node-indexed meshes remove the boundary mismatch introduced by independently adapted slice-pair domains.

The validation connects these numerical results to collective observables. PIC-family and BeamBeam3D round-beam coherent factors lie within 1.42% of the Vlasov reference, the largest same-setting Octopus–BeamBeam3D difference is 0.31%, and the five-slice spectra show similar horizontal structure. The historical 15-slice EIC comparison adds a demanding example of complete multi-turn emittance evolution in both programs. Its nonidentical inputs and incomplete raw provenance make it qualitative application context, not source-pinned validation or a statistical-equivalence study. At EIC production scale, 3.58 million macroparticles and 15 slices advance in about 0.31 s/turn on one workstation GPU, and below 0.2 s/turn on an NVIDIA A100 of the NERSC Perlmutter system. Together, the solver-specific error budget, layered validation, and measured throughput support practical strong–strong studies in the tested domain.

Code and data availability

Octopus is released under the MIT license at `https://github.com/xud929/Octopus.jl` [31]. The source state evaluated for this manuscript is commit 800d74652a79 (`v0.1.11-425-g800d746`, branch `paper-cpc`). Octopus requires Julia 1.12; the CUDA backend uses CUDA.jl 6.2, with FFTW, HDF5/JLD2, and SpecialFunctions as the remaining core dependencies, resolved by the committed project environment (`julia -project=.`). A simulation constructs the two beams and lattice maps and selects the field model by passing one of the four solver objects (`PICPoissonSolver`, `GaussianPoissonSolver`, `SpectralPoissonSolver`, `GaussianPICPoissonSolver`) to the collision element; all four implement the shared `collide!` interface, and `test/examples/strong_strong_tracking.jl` is the worked strong–strong example. The unit and regression test suite runs with `julia -project=. -e 'using Pkg; Pkg.test()'`. BeamBeam3D is cited at the unmodified independent source commit used for the source-pinned comparisons [32]. The Gaussian-subtracted configuration evaluated here is specified in Table 1 and remains experimental. Input decks and selected raw and reduced numerical records, and the reduction and plotting programs underlying the reported figures and numerical tables are provided in the public reproducibility repository `https://github.com/xud929/2026_octopus_cpc`. For the historical EIC comparison, the archive contains BeamBeam3D decks, log, and raw outputs and the reduced Octopus series, but not the native Octopus HDF5 output or the complete original run identities. The supplied Octopus runner at the cited source state enables a new calculation; it does not retrospectively source-identify the plotted historical curves.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

Claude (Anthropic) and Codex (OpenAI) assisted with derivations, implementation, tests, numerical analysis, and editing. The authors checked derivations against limiting cases and source conventions, required executable tests for code changes, recomputed reported values from archived data, and made all scientific decisions. The authors take full responsibility for the software, analysis, and manuscript.

Declaration of competing interest

The authors declare no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Derong Xu: Conceptualization, Methodology, Software, Validation, Formal analysis, Writing – original draft, Writing – review & editing.

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